Ⓑ 2023-J020-1-EL-01/20

Pre-Junior Cycle Final Examination, 2023

Mathematics

Higher Level

Time: 2 hours

270 marks

CANDIDATE DETAILS

DAY and DATE of BIRTH [ ][ ] / [ ][ ] For example, 3rd February is entered as 03/02

NAME

SCHOOL

TEACHER

School Stamp

For Examiner only	
Question	Mark
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
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10.	
11.	
12.	
Total	
%	

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Ⓑ 2023-J020-1-EL-01/20

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 **(Suggested maximum time: 5 minutes)**

5 students estimated the perimeter (in metres) of a model of their new school.
The following table shows their estimates.

Student	Anne	Milo	David	Kamil	Emila
Estimate	$\frac{2\pi}{3}$	$\sqrt{11}-1$	$2^{\frac{3}{2}}$	$\left(\frac{4}{3}\right)^3$	$2{\cdot}119$

(a) Write these estimates in **ascending** order.

(b) The actual perimeter is $2{\cdot}713$ m.
Milo says that his estimate is the most accurate. Is he correct?
Tick (✓) **one** box only. Give a reason for your answer.

Yes ☐ No ☐

Reason:

(c) Express the error in Milo's estimate as a percentage of the actual perimeter, giving your answer correct to one decimal place.

Question 2 **(Suggested maximum time: 10 minutes)**

Jane and Robert both like reading books.
They record the number of pages in each of the books they read.
These are plotted on the stem and leaf plot below.

Jane						**Robert**				
				9	12					
		9	1	1	13	1	3	3	4	
8	7	6	1	0	14	1	2	2	2	3
9	8	3	2	0	15	0	1	2	4	5
		8	3	1	16	0	5	7	9	

Key: | 13 | 4 = 134 pages

(a) Write down the **median** value for Jane and the **median** value for Robert.

Jane:

Robert:

(b) Work out the **mean** value for Jane and the **mean** value for Robert.

Jane:

Robert:

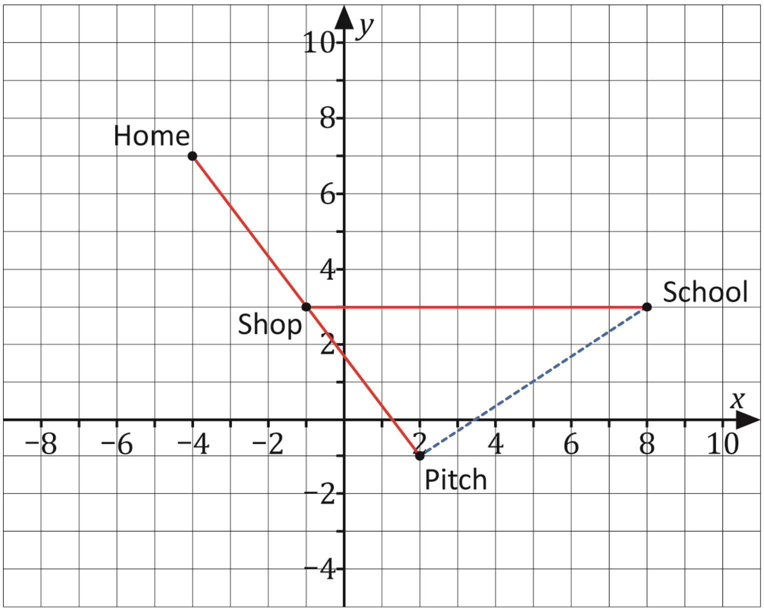
(c) Work out the **range** of values for Jane and the **range** of values for Robert.

Jane:	Robert:

(d) Compare the scores on the results for both Jane and Robert.
Refer to at least one measure of **central tendency** and at least one measure of **variability** (spread) in your answer.

Question 3 **(Suggested maximum time: 15 minutes)**

The co-ordinate diagram below shows the road that Peter travels from his home, past the local shop to the football pitch. Peter also cycles to school along this route, turning left at the junction at the shop.



(a) Peter's home has co-ordinates $(-4, 7)$. Write down the co-ordinates of the school, local shop and football pitch in the table below.

Place	Home	School	Shop	Pitch
Co-ordinates	$(-4, 7)$	(,)	(,)	(,)

(b) Show that the shop is exactly halfway between Peter's home and the football pitch.

(c) Peter cycles to school, taking the road towards the pitch and turning left at the shop. By finding the distance from Peter's home to the shop and the distance from the shop to the school, work out the total distance on the co-ordinate diagram that Peter cycles to school.

(d) The actual distance that Peter cycles to school is $2{\cdot}8$ km. Work out the scale of the map on the co-ordinate diagram.

1 unit = m

(e) A new road is being constructed from the pitch to the school. Show that the slope of this road on the co-ordinate diagram is $\frac{2}{3}$.

(f) Work out the equation of this new road. Give your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.

Question 4 **(Suggested maximum time: 10 minutes)**

(a) A combination lock, **Lock A**, has a 4-digit code.
This code can be made using any digits from 0 to 9 and these digits may be repeated in the code.

An example of a 4-digit code for the combination lock is 0960.

Work out how many different codes are possible for **Lock A**.

(b) Another combination lock, **Lock B**, also has a 4-digit code.
This code can be made using any digits from 0 to 9 but each digit can be used only once.

An example of a 4-digit code for this combination lock is 0961.

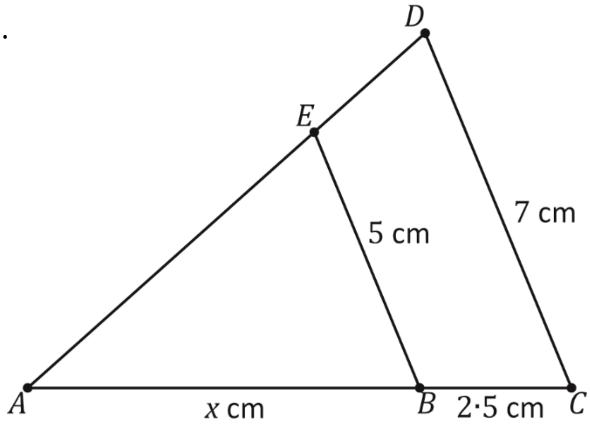
Work out how many different codes are possible for **Lock B**.

(c) Work out the difference between the number of possible codes for **Lock A** and the number of possible codes for **Lock B**.
Express this difference as a percentage of the number of codes possible for Lock A.

(d) Both locks can be purchased in many hardware stores for €28·99. This price includes VAT calculated at 23%. Work out the cost of one lock before VAT is added.
Give your answer correct to the nearest cent.

Question 5 **(Suggested maximum time: 10 minutes)**

(a) The diagram shows a triangle ACD (not to scale).
The line BE is parallel to the line CD.
$|BC| = 2{\cdot}5$ cm, $|BE| = 5$ cm,
$|CD| = 7$ cm and $|AB| = x$ cm.

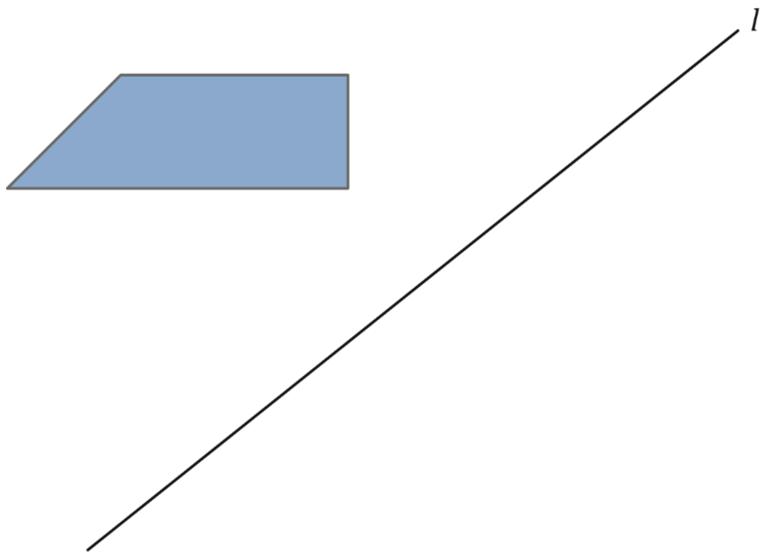


Work out the value of x.

(b) Divide the line segment $[PQ]$, shown below, into **four equal** segments without measuring it.
Show all your construction lines clearly.

Question 6 **(Suggested maximum time: 5 minutes)**

Draw the image of the shape below under S_l, **axial symmetry** in the line l.
Show all your construction lines clearly.



Question 7 **(Suggested maximum time: 10 minutes)**

(a) Factorise fully $18pq - 10rs - 4pr + 45qs$.

(b) Write the following in the form $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{Z}$:

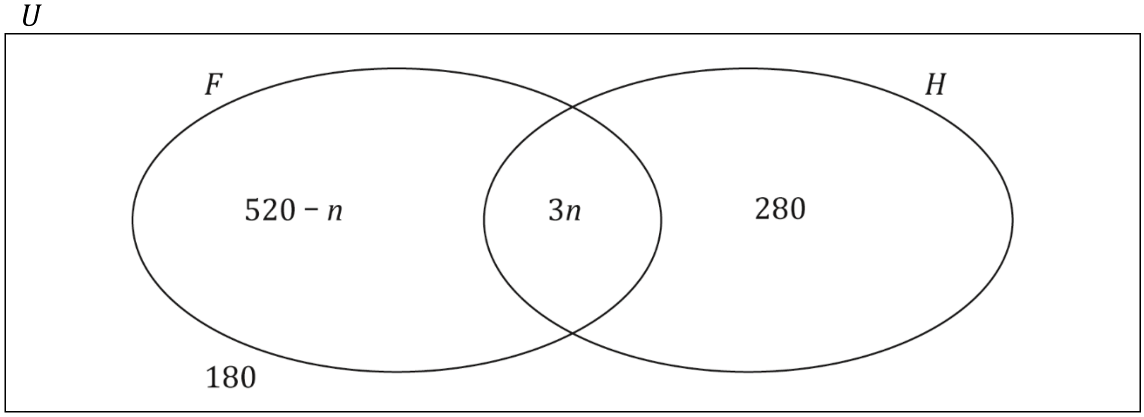
$$2x(x - 5) = 15.$$

(c) Using your answer from part **(b)**, solve the equation $2x(x - 5) = 15$.
Give your answer correct to two decimal places.

(d) $2^{x-1} = 32$. Work out the value of x.

Question 8 **(Suggested maximum time: 5 minutes)**

The Venn diagram below shows the number of members of a GAA club (U) who play football (F) or hurling (H) or both, where $n \in \mathbb{N}$.
Some members do not play either sport.



(a) Write down how many members do not play either sport.

Answer =

(b) Using the Venn diagram, work out the **maximum** number of people who could be members of the GAA club.

Answer =

(c) Using the Venn diagram, work out the **minimum** number of people who could be members of the GAA club.

Answer =

Question 9

(Suggested maximum time: 10 minutes)

Susan commutes for work from Skibbereen to Cork 3 days a week.
The distance from Susan's home in Skibbereen to her place of work is 82 km.
The journey takes an average time of 1 hour and 20 minutes each way.

(a) How long does Susan spend in her car each week commuting to and from work?

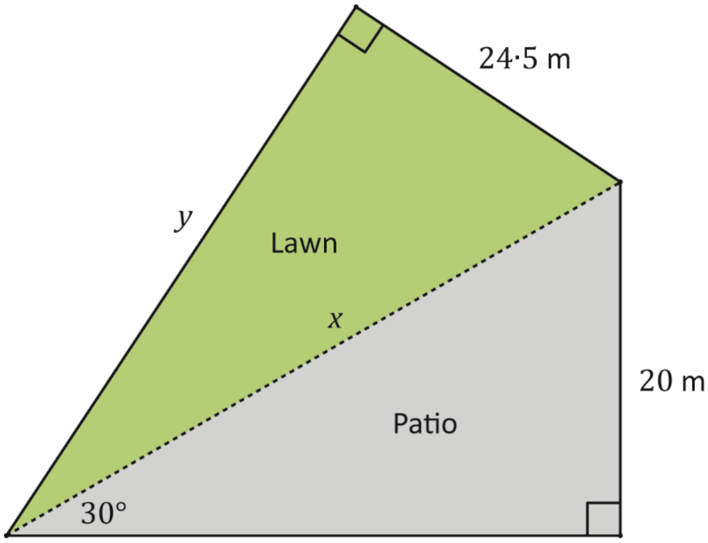
(b) Work out Susan's average speed in km/hour.

(c) Susan says that her car uses petrol at an average rate of $4{\cdot}1$ litres per 100 km.
Work out how much petrol she uses each week commuting to and from work.

(d) Petrol costs Susan $161{\cdot}9$ cent per litre.
Work out how much money Susan saves on petrol each week by working from home for 2 days. Give your answer correct to the nearest euro.

Question 10 **(Suggested maximum time: 10 minutes)**

A garden is made up of a patio area and a lawn area as shown in the diagram.
Both the patio and the lawn form right-angled triangles.



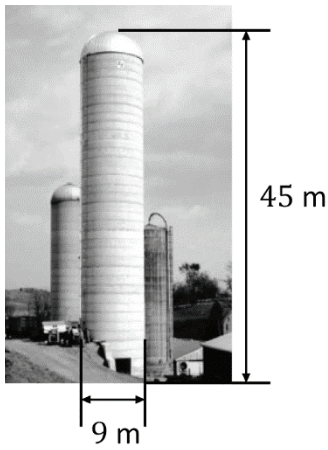
(a) Use **trigonometry** to work out the distance marked x in the diagram.

(b) Use the **Theorem of Pythagoras** to work out the distance marked y in the diagram.
Give your answer correct to one decimal place.

(c) Find the total area of the garden (lawn and patio).
Give your answer correct to the nearest m^2.

Question 11 **(Suggested maximum time: 15 minutes)**

Silos are large narrow structures used on farms for storing grain or silage.
One of the largest silos in the United States is in Pennsylvania.
It is made up of a hemisphere on top of a cylinder.
The diameter of the silo is 9 m and the total height of the structure is 45 m, as shown in the diagram.



(a) Tick (✓) the correct box to show the height of the **cylindrical** part of the silo. Tick **one** box only. Justify your answer.

36 m ☐ 45 m ☐ 40·5 m ☐

Justification:

(b) Work out the volume of the silo.
Give your answer correct to the nearest m^3.

$$V = (\pi \times r^2 \times h) + (\frac{2}{3} \times \pi \times r^3)$$

(c) The capacity of the silo is 2662 m^3.
Give a reason why your answer in part **(b)** is not equal to the stated capacity.

(d) The Swissmill Tower in Zurich is the tallest operating grain silo in the world. At 118 metres high, the tower is in the shape of a rectangular solid with a square base. The tower can store $45\,000$ m^3 of grain.

(i) Work out the width of one side of the tower.
Give your answer correct to one decimal place.

(ii) There is a proposal to cover the tower's exterior with plants as it is considered too industrial-looking. The cost of planting is €250 per m^2.
Work out the cost of covering the four exterior **sides** of the tower with plants.

Question 12 **(Suggested maximum time: 5 minutes)**

(a) Show that the following terms form a **quadratic** pattern.

Term	1	2	3	4	5
Value	3	6	14	27	45

(b) The following terms form a **linear** pattern.
Find the missing values in the table.

Term	1	2	3	4	5
Value	3		–5		–13

(c) Three consecutive numbers add up to 138. N is the smallest of the three numbers.

(i) Write an algebraic expression for the above information.

(ii) Find the value of N and hence write down the three consecutive numbers.

Page for extra work.
Label any extra work clearly with the question number and part.

Q. & part	

Page for extra work.
Label any extra work clearly with the question number and part.

Q. & part	

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Mathematics – Higher Level

Time: 2 hours

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